Quantum Boolean Functions and the KKL Theorem

Sidhant Saraogi

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1 Introduction

While there is no standard generalization of boolean functions to the quantum domain, we present one such generalization, a unitary operator f which squares to identity. We develop a quantum analogue for fourier analysis along with a suitable notion of influence. We summarize arguments for why this definition is a useful generalization by stating a number of interesting and useful results proved in [1]. It is important to note that these generalizations often require novel proof techniques, and these proofs can be used to obtain different proofs for classical theorems since a quantum boolean function reduces to the classical case when f is diagonal in the computational basis.

The Kahn-Kalai-Linial (KKL) theorem [2] states that every unbiased boolean function must have a variable with high influence. We present a possible generalization of the KKL Theorem for quantum boolean functions before proving the generalized theorem for classical boolean functions and for a certain class of anti-commuting boolean functions and examine the conjecture in [1] which says the KKL theorem holds for all quantum boolean functions. We also examine why the classical proof fails to generalize directly to the quantum case. We also present a proof of a quantum Poincaré inequality.

2 Preliminaries

The classical boolean functions take the form $f : \{-1, 1\} \to \{-1, 1\}$. We will denote strings in $\{-1, 1\}^n$ using the boldface $\mathbf{x}$ and strings in $\{0, 1, 2, 3\}^n$ using $\mathbf{s}$. We denote the support of $\mathbf{x} \in \{-1, 1\}^n$ by

$$\mathrm{supp}(\mathbf{x}) := \{j | x_j \neq 1\} \tag{2.1}$$

and the support of $\mathbf{s} \in \{0, 1, 2, 3\}^n$ by

$$\mathrm{supp}(\mathbf{s}) := \{j | s_j \neq 0\} \tag{2.2}$$

When working with quantum boolean functions, we work in the Hilbert space of n qubits, $\mathcal{H} \equiv (\mathbf{C}^2)^{\otimes n}$. We also use standard bra-ket notation for elements

$x \in \{0,1\}^n$ (in place of $\{-1,1\}^n$) when convenient with the inner product $\langle \mathbf{x}|\mathbf{y}\rangle = \delta_{x_1,y_1} \dots \delta_{x_n,y_n}$. We also sometimes work in the Hilbert space $\mathcal{B}(\mathcal{H})$ of bounded operators on $\mathcal{H}$ with the Hilbert-Schmidt inner product $\langle M, N\rangle = \frac{1}{2^n} tr(M^\dagger N)$,
$M, N \in \mathcal{B}(\mathcal{H})$. The normalized Schatten p-norm of a n-dimensional operator f is :

$$\|f\|_p \equiv \left(\frac{1}{n}\sum_{j=1}^{n} s_j(f)^p\right)^{1/p} \tag{2.3}$$

where $\{s_j(f)\}$ are the singular values of f.
Note that if we define $|f| \equiv \sqrt{f^\dagger f}$, we can write the p-norm as,

$$\|f\|_p \equiv \left(\frac{1}{n} tr(|f|^p)\right)^{1/p} \tag{2.4}$$

Note that the p-norm satisfies the equalities $\|f\|_{pq}^p$ and $\|f\|_q^p = \|f^p\|_{q/p}$.

3 Quantum boolean functions

Definition 3.1. A quantum boolean functions is a unitary operator f on n qubits such that $f^2 = \mathbf{I}$.

Since all the eigenvalues of f are ± 1, f is hermitian. Also, every unitary Hermitian operator is quantum boolean. Note that the unitary property of the operator is important since it needs to be implemented on a quantum computer.

The second important property that quantum boolean functions have is that classical boolean functions can be implemented as quantum boolean functions. To see this consider any $f : \{0,1\}^n \to \{0,1\}$. Then two natural ways of implementing f are the bit-oracle [4] where f defines a unitary operator U_f:

$$|\mathbf{x}\rangle|y\rangle \mapsto U_f|\mathbf{x}\rangle|y\rangle := |\mathbf{x}\rangle|y \oplus f(\mathbf{x})\rangle \tag{3.1}$$

and the phase oracle where we also obtain a unitary operator written as f :

$$|\mathbf{x}\rangle \mapsto f(\mathbf{x})|\mathbf{x}\rangle \tag{3.2}$$

also written as

$$|\mathbf{x}\rangle \mapsto f(\mathbf{x})|\mathbf{x}\rangle \tag{3.3}$$

Therefore, the classical boolean functions form a class of examples of quantum boolean functions themselves. Now, we show a way of combining single qubit quantum boolean functions to functions on n qubits.

Definition 3.2. Let f be a quantum boolean function. Then we say that f us local if it can be written as

$$f = U_1 \otimes \dots \otimes U_n \tag{3.4}$$

where U_i is a 2×2 matrix with $U_i^2 = \mathbb{I}$ for all i.

4 Fourier Analysis

In this section, we establish an analogue to fourier analysis for quantum boolean functions. Let the Pauli operators be defined as,

$$\sigma^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

Note that these operators are quantum boolean functions themselves on 1 qubit. We will denote the stabilizer operator $\sigma^{\mathbf{s}} := \sigma^{s_1} \otimes \ldots \otimes \sigma^{s_n}$ where $\mathbf{s} \in \{0,1,2,3\}^n$. Note that the stabilizer operator is simply a tensor product over the pauli matrices. We use the notation $|\mathbf{s}\rangle$ and $\chi_{\mathbf{s}}$ interchangeably to refer to $\sigma^{\mathbf{s}}$. Note that this leads to the following useful inner product equivalence,

$$\langle \mathbf{s}|\mathbf{t}\rangle \equiv \frac{1}{2^n} tr((\sigma^{\mathbf{s}})^\dagger \sigma^{\mathbf{t}}) \equiv \langle \chi_{\mathbf{s}}, \chi_{\mathbf{t}}\rangle. \tag{4.1}$$

Note that the set of stabilizer operators is orthonormal with respect to the Hilbert-Schmidt inner product and therefore forms a basis for $\mathcal{B}(\mathcal{H})$. Therefore, we can express any $f \in \mathcal{B}(\mathcal{H})$ as

$$f = \sum_{\mathbf{s}\in\{0,1,2,3\}^n} \hat{f}_{\mathbf{s}} \chi_{\mathbf{s}}. \tag{4.2}$$

where $\hat{f}_{\mathbf{s}} = \langle \chi_{\mathbf{s}}, f\rangle$. This provides us with a fourier like expansion for f with the $\hat{f}_{\mathbf{s}}$ obviously being the respective fourier coefficients for $\mathbf{s} \in \{0,1,2,3\}^n$. If f is Hermitian, then these coefficients are all real. We define the support of an operator f as

$$\mathrm{supp}(f) \equiv \bigcup_{\mathbf{s}|\hat{f}_{\mathbf{s}}\neq 0} \mathrm{supp}(\mathbf{s}) \tag{4.3}$$

An important result of the defintion of $\{\hat{f}_{\mathbf{s}}\}$ is that when we expand a boolean function f in terms of $\chi_{\mathbf{s}}$, we obtain the classical fourier expansion of f:

Theorem 4.1. *Let $f : \{0,1\}^n \to \{-1,1\}$ be a boolean function. Then if f acts as $f|\mathbf{x}\rangle = f(x)|\mathbf{x}\rangle$, and if*

$$f = \sum_{\mathbf{s}} \hat{f}_{\mathbf{s}} \chi_{\mathbf{s}} \tag{4.4}$$

then the set $\{\hat{f}_{\mathbf{s}}\}$ is given by

$$\hat{f}_{\mathbf{s}} = \begin{cases} 0 & s \notin \{0,3\}^n \\ \frac{1}{2^n} \sum_{T\subset[n]} (-1)^{|S\cup T|} f(T) & \mathbf{s} \in \{0,3\}^n \end{cases} \tag{4.5}$$

where $S = \mathrm{supp}(\mathbf{s})$ and $T = \mathrm{supp}(t)$ where $t_j = 0$ if $j \notin T$ and $t_j = 3$ if $j \in T$.

We also have now the really important quantum analogues of Plancherel's theorem and Parseval's equality,

Theorem 4.2. *Let f and g be operators on n qubits. Then $\langle f, g\rangle = \sum_{\mathbf{s}} \hat{f}^*_{\mathbf{s}} \hat{g}_{\mathbf{s}}$. Also, $\|f\|_2^2 = \sum_{\mathbf{s}} |\hat{f}_{\mathbf{s}}|^2$. Thus, if f is quantum boolean, $\sum_{\mathbf{s}} \hat{f}^2_{\mathbf{s}} = 1$.*

Now we define the notion of fourier weights according to partiuclar levels:

$$f^{=k} := \sum_{\mathbf{s}, |\mathbf{s}|=k} \hat{f}_{\mathbf{s}} \chi_{\mathbf{s}} \tag{4.6}$$

and a definition for the degree of f is

$$\deg(f) := \max_{\mathbf{s}, \hat{f}_{\mathbf{s}} \neq 0} |\mathbf{s}| \tag{4.7}$$

Finally, we note an important difference between and quantum boolean functions. Note that in the classical case every non-zero fourier coefficient must have absolute value atleast $\frac{1}{2^{n-1}}$ but this is not necessary in the quantum case. Consider the operator

$$f := \varepsilon \sigma^1 \otimes \sigma^1 + \sqrt{1-\varepsilon^2} \sigma^2 \otimes \mathbb{I} \tag{4.8}$$

Note that f is a sum of anticommuting boolean functions. By Lemma 4.3, this is a quantum boolean function for any $0 \leq \varepsilon \leq 1$. Taking arbitrary small ε, we see that the fourier coefficient of $\sigma^1 \otimes \sigma^1$ might be arbitrarily small. We include the following lemma from [1] as background for the above example.

Lemma 4.3. *Let $\alpha_j \in \mathbb{R}, j \in [m]$, satisfy $\sum_j \alpha_j^2 = 1$ and $\{f_j\}$, $j \in [m]$, be a set of anticommuting boolean functions, i.e.*

$$\{f_j, f_k\} := f_j f_k + f_k f_j = 2\delta_{jk} \mathbb{I} \tag{4.9}$$

Then

$$f = \sum_{j=1}^{m} \alpha_j f_j \tag{4.10}$$

is a quantum boolean function.

5 Influence

To talk about the KKL Theorem, we first need to establish a notion of influence for quantum boolean functions. We will also state certain important properties for this defintion of the influence and prove a "quantum" Poincaré inequality. We start off by defining some derivative operators :

Definition 5.1. The i-th derivative operator d_i is the superoperator:

$$d_i := \frac{1}{2}(\mathcal{I} - S_i) \tag{5.1}$$

where $\mathcal{I}$ is the identity superoperator and S_i is the spin flip operation on the i-th qubit defined as

$$S(M) = \sigma^2 M^* \sigma^2 \tag{5.2}$$

for any single qubit operator M.
The *gradient* of f is

$$\nabla(f) = (d_1(f), \ldots, d_n(f)), \tag{5.3}$$

and the *laplacian* of f is

$$\nabla^2(f) = \|\nabla(f)\|_2^2 = \sum_{i=1}^{n} \|d_i(f)\|_2^2. \tag{5.4}$$

We immediately get the following lemma from the definition.

Lemma 5.2. *Let f be an operator on n qubits. Then the d_i operator acts as follow.*

$$d_i(f) = \sum_{\mathbf{s}|s_j \neq 0} \hat{f}_{\mathbf{s}} \chi_{\mathbf{s}}. \tag{5.5}$$

We finally get to the definition of influence exactly similar to the classical case

Definition 5.3. Let f be a quantum boolean function. We define the *influence* of the i-th qubit to be

$$I_i(f) := \|d_i(f)\|_2^2 = \sum_{\mathbf{s}|s_j \neq 0} \hat{f}_{\mathbf{s}}^2 \tag{5.6}$$

and the *total influence* to be

$$I(f) := \sum_{i=1}^{n} I_i(f) = \sum_{\mathbf{s}} |\mathbf{s}| \hat{f}_{\mathbf{s}}^2 \tag{5.7}$$

Note that the definition reduces to the classical definition when f is diagonal in the computational basis. Its easy to see that the influence on the i-th qubit measures the change in the value of the function is changed when the state of the i-th qubit is inverted through the origin of the Bloch sphere.

Now, we need to generalize this notion of influence of a set $J \subset [n]$ on f,

Definition 5.4. Let f be a quantum boolean function. Then the influence of a set $J \subset [n]$ on f is

$$I_J(f) := \|d_J(f)\|_2^2, \tag{5.8}$$

where

$$d_J(f) := f - tr_J(f) \otimes \frac{\mathbb{I}}{2^{|J|}} \tag{5.9}$$

The intuition for the above definition comes directly from an alternate characterization of derivatives and influence in the single qubit case provided in **Lemma 68.** of [1]. The above defintion is simply a generalization of the alternate characterization. However, there is no simple intuition for what the influence on multiple qubits denotes.

In attempt to establish a quantum version of the KKL theorem, we first need to establish a notion of variance.

Definition 5.5. Let f be a quantum boolean function. Then the *variance* of f is

$$\mathrm{var}(f) := \frac{1}{2^n} tr(f^2) - \left(\frac{1}{2^n} tr(f)\right)^2 \tag{5.10}$$

Note that this is an obvious generalization of the classical definition of variance.

Theorem 5.6 (Quantum Poincaré Inequality)**.** *Let f be a quantum boolean function on n qubits. Then*

$$var(f) \leq \nabla^2(f) = I(f) \tag{5.11}$$

Proof. Note that

$$\mathrm{var}(f) = \frac{1}{2^n} tr(f^2) - (\frac{1}{2^n} tr(f))^2 = \sum_{\mathbf{s}} \hat{f}_{\mathbf{s}}^2 - \hat{f}_0^2 \leq \sum_{\mathbf{s}} |\mathbf{s}| \hat{f}_{\mathbf{s}}^2 = I(f), \tag{5.12}$$

as required. □

The following simple corollary of the above theorem follows similarly from the proof in the classical case

Corollary 5.7. *Let f be an unbiased quantum boolean function i.e. $tr(f) = 0$. Then there exists $j \in [n]$ such that $I_j(f) \geq 1/n$.*

6 Quantum KKL Theorem

The KKL Theorem [2] states that for every unbiased boolean functions $f : \{0,1\}^n \to \{-1,1\}$, there exists a $j \in [n]$ such that $I_j(f) \geq \mathrm{var}(f)\Omega(\frac{\log n}{n})$. Note that we have proved in Theorem 5.6 that every unbiased quantum boolean function on n qubits, there is a qubit j such that $I_j(f) \geq 1/n$. We therefore have the following conjecture for a quantum analogue of the KKL Theorem.

Conjecture 6.1. *Let f be a quantum boolean function on n qubits. Then there exists $j \in [n]$ such that $I_j(f) \geq \mathrm{var}(f)\Omega(\frac{\log n}{n})$.*

It is important to note that the classical proof does not simply generalize to the quantum case. Note that in the classical case, if the derivative operator in a particular direction has low norm, then it has small support, implying that it has some large high degree fourier coefficient, which must then be included in derivatives in many directions. This leads to the proof that if all influences are small, then their sum is large which is used to obtain the proof for the KKL theorem. However, in the quantum case, this is not true as there exist quantum boolean functions whose derivative is small in a particular direction, but which are also low degree. This case is illustrated by the example in (4.8). On level 2, the only fourier coefficient is ϵ which can be arbitrary small (hence the function is low degree), and also the derivative $d_2(f) = \epsilon\sigma^1 \otimes \sigma^1$. Then $\|d_2(f)\|_2 = \epsilon$ which can also be arbitrarily small.

Now we examine a specific proof of the classical KKL Theorem and aim to demonstrate why it fails in the quantum case. We prove first the following quantum generalization of a theorem due to Talgrand [3]. To do this, we first need a quantum version of a hypercontractivity result. We state without proof the following theorem (originally a corollary) from [1]:

Theorem 6.2. *Let f be a Hermitian operator on n qubits with degree atmost d. Then, for any $q \geq 2$, $\|f\|_q \leq (q-1)^{d/2}\|f\|_2$. Also, for any $p \leq 2$, $\|f\|_p \geq (p-1)^{d/2}\|f\|_2$.*

Theorem 6.3. *Let f be a traceless (unbiased) Hermitian operator on n qubits. Then*

$$\|f\|_2^2 \leq \sum_{i=1}^{n} \frac{10\|d_i f\|_2^2}{(2/3)\log\left(\|d_i f\|_2/\|d_i f\|_1\right) + 1} \tag{6.1}$$

Proof. Let, for any operator g,

$$M^2(g) = \sum_{\mathbf{s}\neq 0} \frac{\hat{g}_{\mathbf{s}}^2}{|\mathbf{s}|} \tag{6.2}$$

Note then that

$$\|f\|_2^2 = \sum_{i=1}^{n} M^2(d_i(f)) \tag{6.3}$$

Now consider for some integer $k \geq 0$,

$$
\begin{aligned}
M^2(g) &= \sum_{\leq |\mathbf{s}| \leq k} \frac{\hat{g}_{\mathbf{s}}^2}{|\mathbf{s}|} + \sum_{|\mathbf{s}|>k} \frac{\hat{g}_{\mathbf{s}}^2}{|\mathbf{s}|} \\
&\leq \sum_{i=1}^{k} \frac{2^i}{i} \|g^{=i}\|_{3/2}^2 + \frac{1}{k+1} \sum_{|\mathbf{s}|>k} \hat{g}_{\mathbf{s}}^2 && \text{(Hypercontractive Inequality)} \\
&\leq \|g\|_{3/2}^2 \sum_{i=1}^{k} \frac{2^i}{i} + \frac{1}{k+1} \|g\|_2^2. && (\|g^{=i}\|_{3/2}^2 \leq \|g\|_{3/2}^2 \text{and} \sum_{|\mathbf{s}|>k} \hat{g}_{\mathbf{s}}^2 \leq \|g\|_2^2) \\
&\leq \frac{1}{k+1} (2^{k+2} \|g\|_{3/2}^2 + \|g\|_2^2) && (\text{since} \sum_{i=1}^{k} \frac{2^i}{i} \leq \frac{2^{k+2}}{k+1})
\end{aligned}
$$

We pick a suitable m such that m is the smallest integer for which $2^{m+1}\|g\|_{3/2}^2 \leq \|g\|_2^2$ and $m \geq 0$. Then $m+1 \geq \frac{1}{2}(2\log(\frac{\|g\|_2}{\|g\|_{3/2}} + 1)$. Finally we obtain that $M^2(g) \leq \frac{1}{k+1}(2^{k+2}\|g\|_{3/2}^2 + \|g\|_2^2) \leq \frac{10\|g\|_2^2}{(2/3)\log(\|g\|_2/\|g\|_{3/2}+1)} \leq \frac{10\|g\|_2^2}{2\log(\|g\|_2/\|g\|_1)+1}$. where the last inequality follows since $(\|g\|_2/\|g\|_{3/2})^3 \geq \|g\|_2/\|g\|_1$ by Cauchy-Schwartz. Substituting g with the $d_i f$'s and summing over i, we get the inequality. □

While for the classical case, the above inequality can be used with the fact that $\{d_i\}$ functions take values in $\{0, \pm 1\}$ to obtain a proof of the KKL theorem, there is no such restriction on the eigenvalues of the $\{d_i f\}$ operators in the quantum case. Therefore, the classicaly proof does not generalize straightforwardly to the quantum case .

Note that in Talgrand's inequality, the worst case scenario is when 2-norms and 1-norms of the operators $\{d_i f\}$ are the same. This situation provides with one extreme case when trying to adapt a classical proof of the KKL theorem for the quantum case. In this section, we start off by showing that in this case, f must be a sum of anticommuting boolean functions.

Definition 6.4. Let f be a quantum boolean function on n qubits. Then the set $J \subset [n]$ is said to have *bad influence* on f if

$$
\left\| f - tr_J(f) \otimes \frac{\mathbb{I}}{2^{|J|}} \right\|_2 = \left\| f - tr_J(f) \otimes \frac{\mathbb{I}}{2^{|J|}} \right\|_1 \tag{6.4}
$$

Lemma 6.5. *Let M be an $n \times n$ Hermitian matrix. If $\|M\|_2 = \|M\|_1$ then the eigenvalues λ_j of M satisfy*

$$
\lambda_j \in \{-\alpha, +\alpha\}, j \in [n], \tag{6.5}
$$

for some $\alpha \in \mathbb{R}$.

Now, we are equipped to prove that quantum boolean functions with bad influence must be sums of anticommuting boolean functions :

Theorem 6.6. *Let f be a quantum boolean function on n qubits. Then $J \subset [n]$ has bad influence on f iff*

$$f = \sqrt{1-\alpha^2} f' \otimes \mathbb{I}_J + \alpha g \tag{6.6}$$

where f' and g are quantum boolean functions, $\{f' \otimes \mathbb{I}_J, g\} = 0$, and $\alpha^2 = I_J(f)$.

Proof. Note that we can write,

$$f = f_{J^c} \otimes \mathbb{I}_J + \sum_{supp(\mathbf{s}) \subset J, \mathbf{s} \neq 0} f_{\mathbf{s}} \otimes \chi_{\mathbf{s}}. \tag{6.7}$$

Then the derivative with respect to J is

$$d_J(f) = \sum_{supp(\mathbf{s}) \subset J, \mathbf{s} \neq 0} f_{\mathbf{s}} \otimes \chi_{\mathbf{s}}. \tag{6.8}$$

Since J, has bad influence by our previous theorem we have that

$$d_J(f) = \alpha \mathbb{I}, \tag{6.9}$$

Suppose we have that $g = \frac{1}{\alpha} d_J(f)$, then $g^2 = \mathbb{I}$. Then $f = f_{J^c} \otimes \mathbb{I}_J + \alpha g$. Note that if we square f, we get that $\mathbb{I} = f_{J^c}^2 \otimes \mathbb{I}_J + \alpha^2 \mathbb{I} + \alpha \{f_{J^c} \otimes \mathbb{I}_J, g\}$ which implies that $\{f_{J^c} \otimes \mathbb{I}_J, g\} = 0$. Note that now if we define

$$f' = \frac{1}{\sqrt{1-\alpha^2}} f_{J^c} \tag{6.10}$$

then we have that $f'^2 = \mathbb{I}$, and we are done. □

Let f be a quantum boolean funciton with $f = \sum_s \hat{f}_{\mathbf{s}} \chi_{\mathbf{s}}$. If $[\chi_{\mathbf{s}}, \chi_t] = 0$ for all $s \neq t$ where $\hat{f}_{\mathbf{s}}, \hat{f}_t \neq 0$, then we call f commuting. If $\{\chi_{\mathbf{s}}, \chi_t\} = 0$ for all $s \neq t$ where $\hat{f}_{\mathbf{s}}, \hat{f}_t \neq 0$, then we call f anti-commuting.Note the classical KKL theorem gives us that commuting quantum boolean functions that can be diagonalised by local unitaries have a qubit with influence atleast $\Omega(\frac{\log n}{n})$. This follows because the influence of each qubit over the function is invariant under local unitary operations. We now present perhaps the most compelling evidence for the quantum KKL theorem holding for arbitary quantum boolean functions. We prove that anti commuting boolean functions have a influential qubit, in effect proving a stronger version of the KKL theorem for such functions.

Theorem 6.7. *Let f be an anticommuting boolean function on n qubits. Then there exists $j \in [n]$ such that $I_j(f) \geq \frac{1}{\sqrt{n}}$.*

Proof. Let $f = \sum_{i=1}^m \hat{f}_{\mathbf{s}_i} \sigma^{\mathbf{s}_i}$ for some m where the coefficients are real. Let S_j be the set of indices of $\{\sigma^{\mathbf{s}_i}\}$ that act non-trivially on qubit j. Note that this is

the same as the set $\{k : d_j(f_k) \neq 0\}$. Then

$$\sum_{j=1}^{n} I_j(f)^2 = \sum_{j=1}^{n} (\sum_{i \in S_j} \hat{f}_{\mathbf{s}_i})^2$$

$$\text{(since } I_j(f) = \sum_{i \in S_j} \hat{f}_{\mathbf{s}_i})$$

$$= \sum_{j=1}^{n} \sum_{i,k \in S_j} \hat{f}^2_{\mathbf{s}_i} \hat{f}^2_{\mathbf{s}_k}$$

$$= \sum_{i,k=1}^{m} \hat{f}^2_{\mathbf{s}_i} \hat{f}^2_{\mathbf{s}_k} |P_{i,k}|$$

$$\text{(by rearranging the sum where } P_{i,k} = \{j | i, k \in S_j\})$$

$$\geq \sum_{i,k=1}^{m} \hat{f}^2_{\mathbf{s}_i} \hat{f}^2_{\mathbf{s}_k} = (\sum_{i=1}^{m} \hat{f}^2_{\mathbf{s}_i})^2 = 1.$$

(since $|P_{i,k}| \geq 1$ as for $\hat{f}_{\mathbf{s}_i}, \hat{f}_{\mathbf{s}_k}$ to anticommute, they must both act non-trivially on a particular qubit.)

Now that we have that $\sum_{j=1}^{n} I_j(f) \geq 1$, it follows simply that there exists $j \in [n]$ such that $I_j(f) \geq 1/\sqrt{n}$. □

7 Conclusion

Note that since we have a quantum KKL theorem type proof for both the "best" case with respect to the inequality i.e. for the classical boolean functions and for the "worst" case, that of the anticommuting boolean functions, it is reasonable to conjecture that the KKL theorem might hold for all quantum boolean functions. If that is the case, or otherwise, it is also useful to examine a quantum analogue of Friedgut's Theorem, the proof of which is based on similar ideas. Friedgut's Theorem [5] for classical boolean functions is states that functions with low average sensitivity depend on few co-ordinates.

References

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